Overview of Cloud Computing

Task 1

Sign up for a free AWS or Azure account and log in to the Management Console. Take a screenshot of your dashboard showing your name or email in the top bar.

Steps followed:

1. Went to the AWS official website (aws.amazon.com) and clicked on "Create an AWS Account".

2. Entered a valid email address, set an account name, and created a strong password.

3. Verified the email address using the OTP/confirmation link sent by AWS.

4. Filled in contact details (name, phone number, and address) and selected the account type as "Personal".

5. Added payment details (card) for identity verification, as required by AWS for the Free Tier.

6. Completed phone number verification through an OTP call/SMS.

7. Selected the Basic (Free) support plan and completed the sign-up process.

8. Logged in to the AWS Management Console using the root user email and password.

9. Confirmed that the account name/email was visible in the top navigation bar of the console, confirming successful login.

Task 2

In the AWS or Azure Management Console, locate and list 5 key services (such as Compute, Storage, Database, Networking, Analytics). For each, write one line describing what it does and name one real app (like Instagram, Zomato, Flipkart) that could use it.

Service	Description	Real App Example
EC2 (Compute)	Provides resizable virtual servers in the cloud to run applications and handle processing workloads.	Flipkart
S3 (Storage)	Offers scalable object storage to store and retrieve any amount of data such as images, videos, and backups.	Instagram
RDS (Database)	Managed relational database service used to store, organise, and query structured application data.	Zomato
VPC (Networking)	Lets users create an isolated private network in the cloud to control traffic flow and security.	Paytm
CloudWatch (Analytics/Monitoring)	Monitors application performance, collects logs, and generates metrics to track system health.	Swiggy

Task 3

Use the search bar in your AWS or Azure Console to find the S3 (AWS) or Blob Storage (Azure) service. Write down the steps you took to locate it and describe in 2-3 lines how a music app like Spotify could use this service to store user-uploaded files.

Steps followed:

1. Logged in to the AWS Management Console.

2. Clicked on the search bar at the top of the console (labelled "Search for services, features, blogs, docs, and more").

3. Typed "S3" in the search box.

4. Selected "S3" from the list of suggested services that appeared in the dropdown.

5. Landed on the S3 dashboard, which displayed the option to create and manage storage buckets.

How Spotify could use S3:

A music streaming app like Spotify can use S3 to store large volumes of audio files, album art, and playlist data uploaded by users or artists. Since S3 offers virtually unlimited and durable storage, Spotify does not need to worry about running out of space as more songs and users are added. It can also use S3 to quickly retrieve and stream files to millions of users at once with high reliability.

Task 4

Pick one key characteristic of cloud computing (on-demand self-service, broad network access, resource pooling, rapid elasticity, or measured service) and explain, in your own words, how it benefits users of a popular app you use daily (e.g., Paytm, BookMyShow, IRCTC). Hint: Think about how these apps handle sudden spikes in users or data.

Characteristic chosen: Rapid Elasticity

Rapid elasticity means that cloud computing resources can be automatically scaled up or down depending on demand, without any manual effort from the user or the company. This is very useful for an app like BookMyShow, which faces huge spikes in traffic whenever tickets for a popular movie or concert go on sale. During normal days, the app may only need a small number of servers to function smoothly, but the moment ticket booking opens for a big release, millions of users try to log in and book seats within minutes.

Because of rapid elasticity, BookMyShow's cloud infrastructure can automatically add extra servers and computing power to handle this sudden rush, preventing the app from crashing or slowing down. Once the rush is over and traffic returns to normal, the extra resources are released automatically, so the company does not keep paying for capacity it does not need. This directly benefits users because they get a fast, reliable booking experience even during high-demand events, without having to worry about server errors or failed transactions.